



Bahir Dar Institute of Technology (BiT)

Faculty of Mechanical and Industrial Engineering

Laboratory Manual
Strength of Materials I

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1. Laboratory Report Writing

Effective teaching and learning of engineering involve seeing, handling, and manipulating real objects and materials. Engineering lab equipment allows students to interact directly with the data gathered and develop skills in a practical investigation. The students would get a first-hand learning experience by performing various experiments on their own. The presentation and report of any engineering or scientific work are probably as important as the actual work. Credit for discovery or development of new concepts cannot be received until the work has been described in a report or paper that is readily available for other people to read. Engineering laboratory reports must be written in a formal way from starting the experiment up to the final result. Therefore, the general structure of the report is:

1. Title
2. Objective
3. Theory
4. Apparatus
5. Procedure
6. Result
7. Calculations
8. Conclusion and discussion

1. Title

They are usually written:

- Name of the institute
- Name of the Faculty/school/ department
- Title of the experiment
- Name(s) of the students(s)
- Date of experiment
- No abbreviations.

2. Objectives

This is a brief statement of what you set out to determine, investigates, test or confirm.

3. Theory

For an engineering laboratory class, this will properly involve one or two explanations, and the statement of key mathematical expressions.

4. Apparatus

Alternative titles are equipment or experimental installations. In this section diagrams and words shall be imparted.

5. Procedure

This is the section in which you describe how you conducted the experiment. There is a well-established tradition of using the passive voice when describing an experimental procedure. In the passive voice, the object of a normal sentence (active voice) becomes the subject. So, describe your procedure using a ‘we did that’ or ‘I did that’ style. You will probably need to use the passive voice, but not for everything. Avoid the clumsier passive voice construction. Finally, keep your descriptions simple; don’t write more than is needed.

6. Result

If your experiment gave a reasonable number of results, it is probably best to present them in the form of a table. You must distinguish values that were actually measured and values that were calculated from the measurement. Remember to place a clear title at the top of each column defining the parameter, giving the symbol and units.

7. Calculation

Calculation of the relationship between parameters likely to involve plotting graphs, interpretation of graph, and perhaps comparisons with theory, will usually be the ‘aim’ of the experiment.

8. Discussion and Conclusion

The real purpose of laboratory classes is to allow you to learn something, so perhaps the conclusion should start ‘I learned the following ...’ but in your report, you have stated an aim (‘to determine...’, to investigate...,) and the conclusion should relate this. You should summarize the outcome of your experiment, what you have determined, or what the investigation has shown.

2. General Safety Guidelines

The following are a list of general safety rules based on the occupational Safety and Health Administration (OSHA), to follow while working on the workshop. Although this is not a list of all requirements, following these rules may prevent you or a coworker from becoming involved in an incident.

- i. **Wear proper clothing for the type of work being done.** Do not wear loose-fitting clothes or jewelry that can get caught in moving parts. Do roll up long sleeves. Do not wear highly flammable clothes. Do not wear sandals, open-toed or canvas shoes. Wearing safety toed shoes are preferable for protection of feet and toes.
- ii. **Remove** all jewelry, including rings, necklaces, bracelets, and watches that could get caught in tooling before starting work.
- iii. **Safety Glasses** must be worn at all times in the shop. Some operations may require additional eye protection or other personal protective equipment
- iv. **Hair.** Pull back long hair in a band and tuck it under your shirt collar or a cap to keep it from getting caught in tools. Long beards must also be contained.
- v. **Keep the work area clean.** Keep the floor free of scraps and oil. Cluttered work areas invite accidents. Keeping workshop and storage spaces clean and dry can help prevent many accidents. Sparks can ignite scraps, sawdust, and solvents. Water can conduct electricity. Do not stand in water, on damp floors or in the rain when working with electrical tools. Keep hands and tools dry.
- vi. Using of **food and drinks** are forbidden in the laboratory.
- vii. **Cell phones** and other personal electronic devices must not be used with working at any apparatus or machine. Loud music is prohibited.
- viii. **Use the correct tools for the job.** Do not use a tool or attachment for something it was not designed to do. Select the correct bit, cutter or grinding wheel for the material with which you are working. This saves time and improves the quality of work and reduces the risk of mishap. If necessary, consult the instructions or shop manual, or call a dealer or an expert on tool use.
- ix. **Know the machine.** Before using any tool, read the operator's manual, or comparable literature as available, to learn the applications, limitations, and potential of each power

tool. Never use a tool unless trained to do so. Inspect it before each use and replace or repair if parts are worn or damaged. Repair tools only if you are trained to do so. Inspect screws, nuts, bolts, and movable parts to make sure they are tightened. Make sure the cord will not become caught or tangled. The cord should be flexible, but not easy to knot. Clean the cord regularly and inspect the grounding connections. Use a ground fault circuit interrupter when working with power tools.

- x. **Know the switch location(s).** Remember where the switch is located so you can turn off the machine quickly.
- xi. **Fires.** Shop fires can be any, or all, of three fire classes: Class A, ordinary combustibles; Class B, combustible liquids; and Class C, fires in live electrical equipment. To eliminate the need for extinguishers for all three classes, install at least one 20-pound ABC Class extinguisher in a convenient location, and possibly more, depending on the size of the shop.
- xii. **Storage.** Develop a system of racks, bins and tool panels to make it easy to find the right tool or materials quickly. Don't store tools, supplies or spare parts in the aisle or on the floor where they become tripping hazards. Keep other flammable materials away from heaters and welding areas to prevent fire. Grease, oil, paint, and solvents should be stored in a closed metal container, preferably in metal cabinets. Gasoline or other fuels should never be stored inside the shop. Supplies and equipment should be stored in an area designed specifically for them.
- xiii. **Do not use** damaged equipment or equipment that does not appear to be operating normally.
- xiv. **Immediately report** any problems or concerns.

3. Shear and Bending of Cantilever Beam

3.1. Introduction

The shearing force and bending moment device have been designed to show that in a beam subjected to transverse loads, at any section of the beam:

- The shearing force is defined as the algebraic sum of the transverse components of the forces to one side of the section.
- The bending moment is defined as the algebraic sum of the moments of the forces at one side of the section.

Several simple experiments will be carried out in order to determine the magnitudes of these moments and these bends while observing the bending process in a beam.

3.2. Background

Theoretical bases

Considering the following cantilever beam subjected to a certain number of transverse loads:

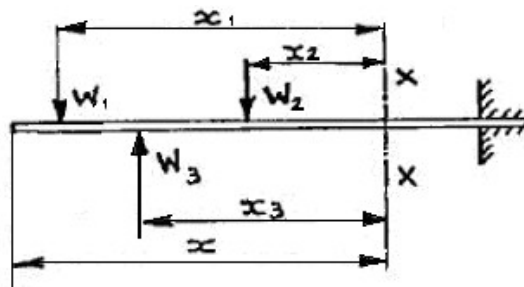


Figure 1: load application distance

For any transverse section xx we will have:

1 shearing force

$$Q = W_3 - W_2 - W_1$$

or...

$$Q = \sum W$$

2 bending moment

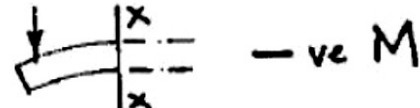
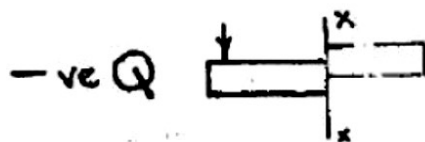
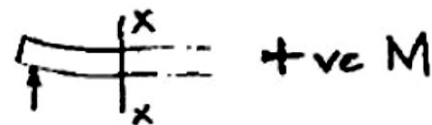
$$M = W_3 x_3 - W_2 x_2 - W_1 x_1$$

or...

$$M = \sum Wx$$

Note-sign convention

Upward forces to the left of any section are considered as positive forces and downward forces as negative forces, producing positive or negative values of Q and M as shown below:



Since Q and M will vary along the length of the beam, when considering beam problems, it is convenient to plot a series of values for Q and M in the form of graphs. These graphs are usually referred to as shearing force and bending moment diagrams.

Example: Cantilever beam with concentrated end load

Let the origin of the graph be at the left-hand end of the beam so that all values of x are positive.

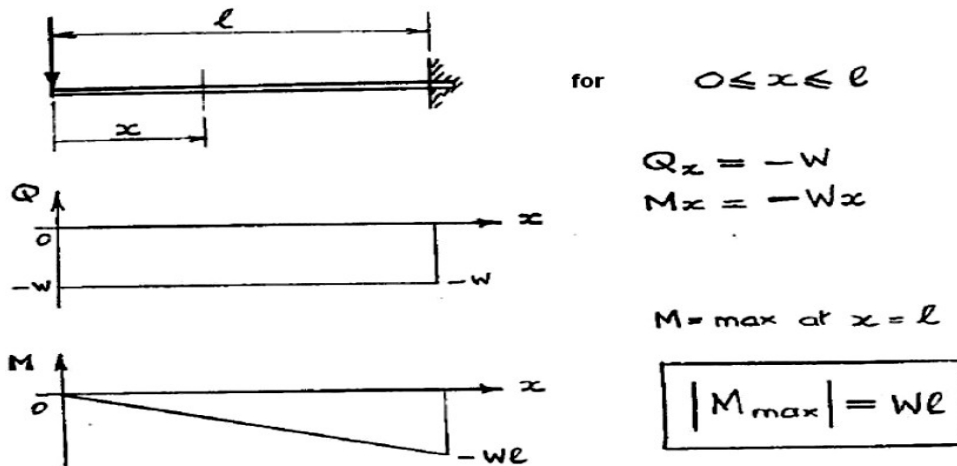
**3.3. Apparatus Description**

Figure 2: shear and bending unit

The shearing force and bending moment device is formed by a structure of steel and aluminum that makes it possible to perform a mechanical study of different structures.

An articulated structure makes it possible to convert force into a bending moment and a shearing force on the beam, with a vertical component. Both effects are compensated by the use of a set of weights that balance the system due to a double-pulley system.

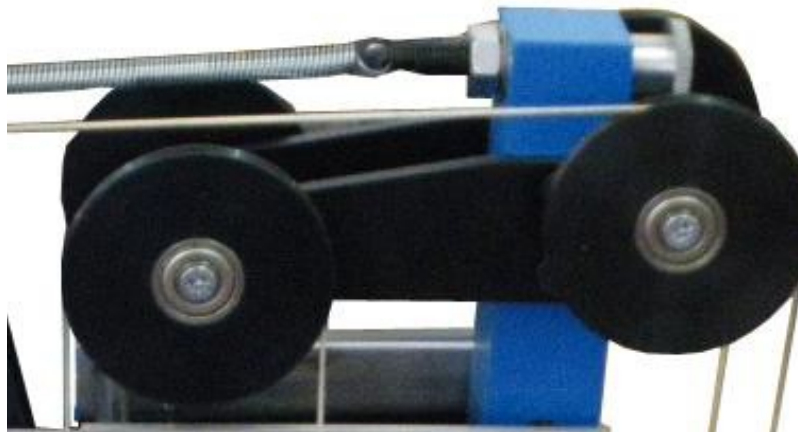


Figure 3: double pulley system

The different points in which the force can be applied make it possible to study the different reactions of the beam, allowing us to compare the different results.

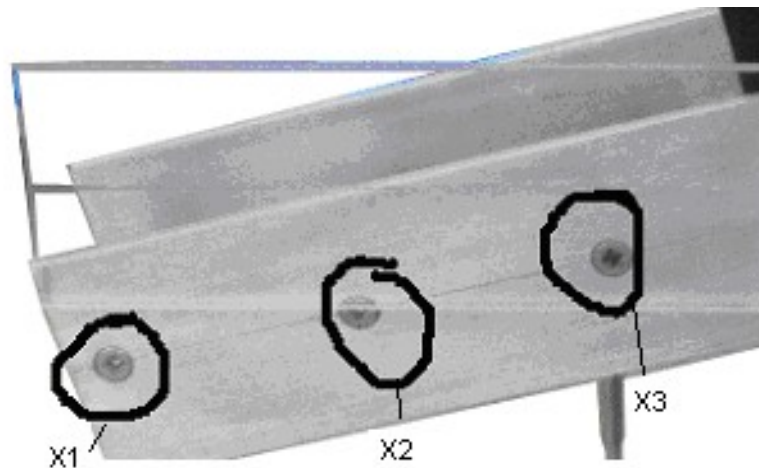


Figure 4: load application points

Specifications

- ❖ Structure made of steel and aluminum.
- ❖ Diagram of the system:

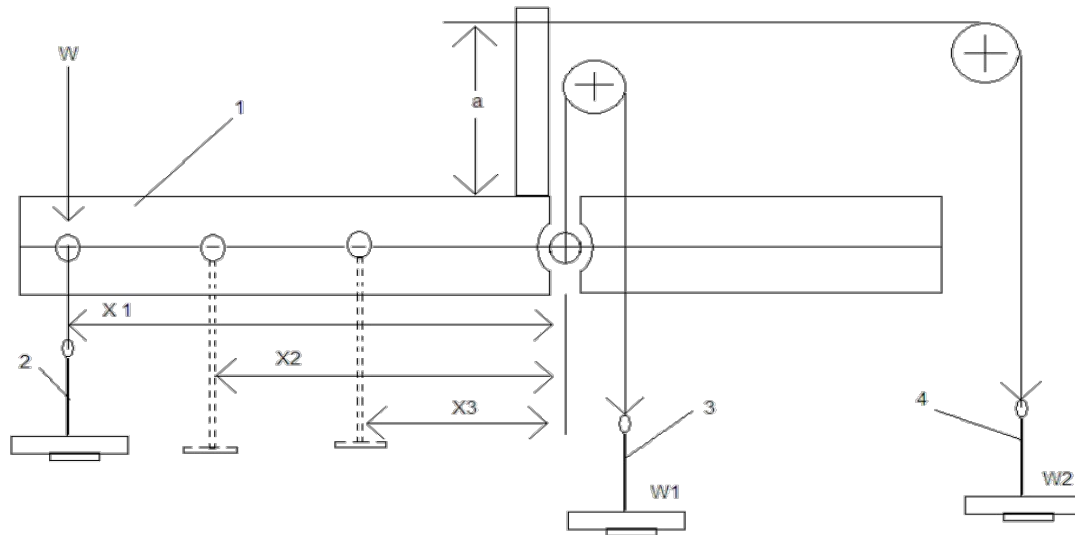


Figure 5: system diagram

$$X1 = 0.15 \text{ m} = 15 \text{ cm} \quad X3 = 0.05 \text{ m} = 05 \text{ cm}$$

$$a = 0.10 \text{ m} = 10 \text{ cm} \quad X2 = 0.10 \text{ m} = 10 \text{ cm}$$

3.1.4 Weights and dimensions

- ❖ Approximate dimensions of the device: 400x200x300 mm
- ❖ Approximate volume of package: 0.16m³
- ❖ Approximate weight: 5 kg.

Services required

- ❖ No particular service.

3.4. Laboratory Practical Exercises

The following figure offers a description of the MMF device, in which you will see the measurements necessary for the performance of the practices.

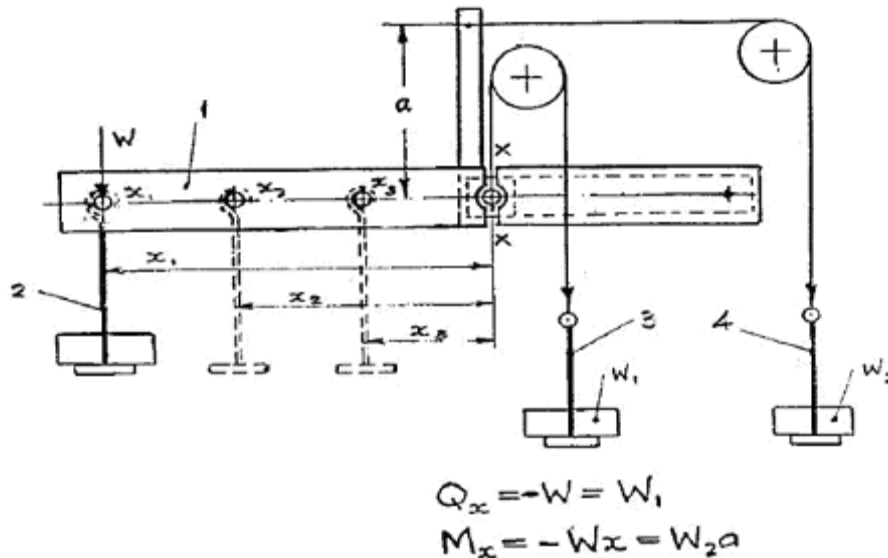


Figure 6: system diagram

Where:

$$X1 = 0,15\text{m}$$

$$X2 = 0,10\text{m}$$

$$X3 = 0,05\text{m}$$

3.3.1 Practical exercise 1:

Study of the shearing force and the bending moment in a cantilever beam

3.3.1.1 Necessary items

In order to apply the force a set of weights will be necessary.

3.3.1.2 Experimental procedure

Step 1: Without there being any load applied to the beam, check that the beam is in its equilibrium position. Adjust the tension springs if necessary. Its equilibrium position is that in which the mark on the transparent screen coincides with the straight mark on the beam.

Step 2: Attach the load hanger (2) and weights to the beam at position X1 and add weights to the hangers (3) and (4) until the beam returns to its equilibrium position.

Step 3: Repeat the experiment with different loads for the previous position, balancing hangers (3) and (4) with weights.

Step 4: Repeat the two previous steps, this time applying loads with the load support (2) but in positions X2 and X3 and weights in positions (3) and (4).

Record the values for each position of the load W and draw the shearing force and bending moment diagram.

Table 1: Inputs and results

No.	App Point	Load applied (W)	W1 (3) grams	W2 (4) grams

The shearing force will be:

$$Q = W1$$

In order calculate the bending moment in that section you will have to consider the following:

$$M = W2 \cdot a$$

4. Deflection of beam

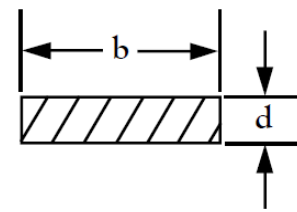
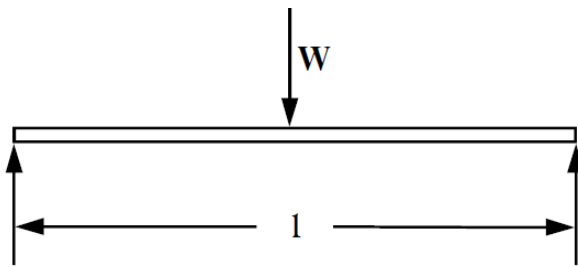
4.1. Theory

The apparatus has been designed to enable students to carry out experiments on simply supported and cantilever beams in order to investigate:

- The relationship between the deflections and the applied loads.
- The effect of variations in length and cross-sectional dimensions on the beam compliance.

i.e.: Deflections per unit load.

4.1.1. Simply Supported Beam with Central Point Load



Beam Cross Section

For this arrangement, it can be shown that the deflection under the load,

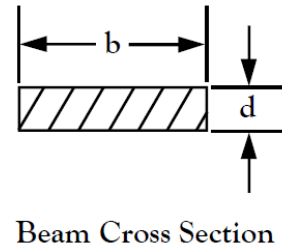
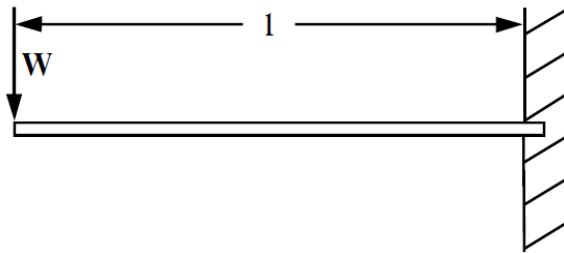
i.e.: - Maximum Deflection

$$\Delta = \frac{Wl^3}{48EI}$$

$$I = \frac{bd^3}{12}$$

$$\therefore \text{Beam compliance} = \frac{\Delta}{W} = \frac{l^3}{4Ebd^3}$$

4.1.2. Cantilever Beam with End Point Load



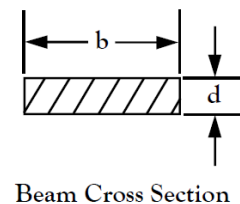
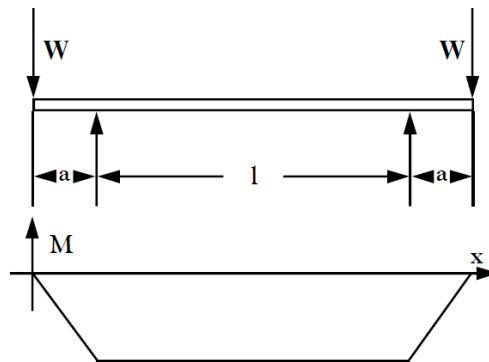
For this arrangement, it can be shown that the deflection under the load,

i. Maximum Deflection

$$\Delta = \frac{Wl^3}{3EI} \quad \text{where } I = \frac{bd^3}{12}$$

$$\therefore \text{beam compliance} = \frac{\Delta}{W} = \frac{4l^3}{Ebd^3}$$

4.1.3. Simply Supported Beams Subjected to Uniform



Bending Moment Diagram

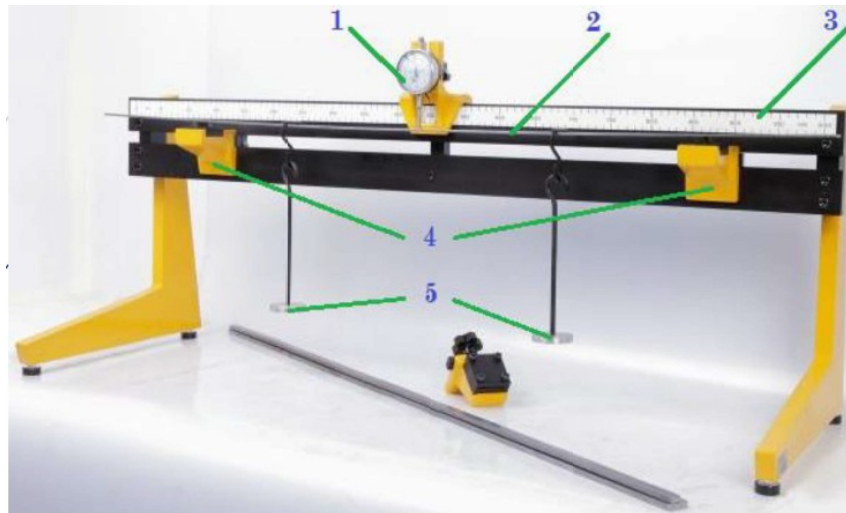
For this arrangement, it can be shown that the central deflection relative to the supports i.e:—
Maximum Deflection between the supports

$$\Delta = \frac{W a I^2}{8EI} \quad \text{where } I = \frac{bd^3}{12}$$

$$\therefore \text{beam compliance } \Delta = \frac{W}{2Eb d^3} = 3aI^2$$

$$W = 2Eb d^3$$

4.2. Description of equipment



1. Dial indicator
2. Specimen
3. Length indicator
4. Free supporter
5. Load hanger

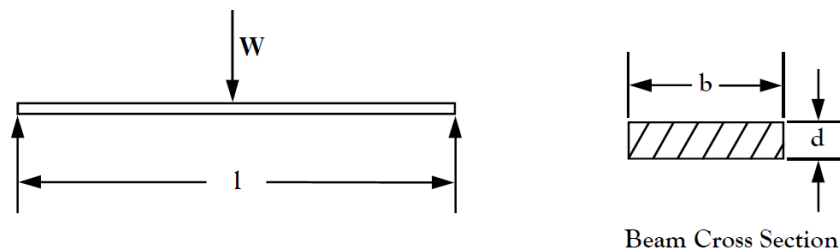
Figure 7: beam deflection equipment

The following are suggestions for a series of experiments that may be carried out on the standard equipment.

4.2.1. Experiment 1

To investigate, for a simply supported beam carrying a central point load,

- (i) The relationships between the deflections and the applied loads.
- (ii) The effect of variations in length and cross-sectional dimensions on the beam compliance.



Carry out load /deflection tests, for beams of the same material as follows:

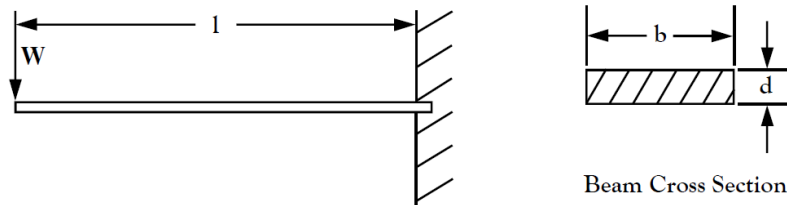
1. Test (i) $l = 800\text{mm}$ $b = 25.4\text{mm}$ $d = 3.2\text{mm}$
2. Test (ii) $l = 400\text{mm}$ $b = 25.4\text{mm}$ $d = 3.2\text{mm}$
3. Test (iii) $l = 800\text{mm}$ $b = 12.7\text{mm}$ $d = 6.4\text{mm}$
4. Test (iv) $l = 800\text{mm}$ $b = 25.4\text{mm}$ $d = 6.4\text{mm}$

- ✓ For the effect of decreasing l compare tests (i) and (ii)
- ✓ For the effect of increasing b compare tests (iii) and (iv)
- ✓ For the effect of increasing d compare tests (i) and (iv)

4.2.2. Experiment 2

To investigate, for a cantilever beam carrying an end point load,

- (i) The relationships between the deflections and the applied loads.
- (ii) The effect of variations in length and cross-sectional dimensions on the beam compliance.



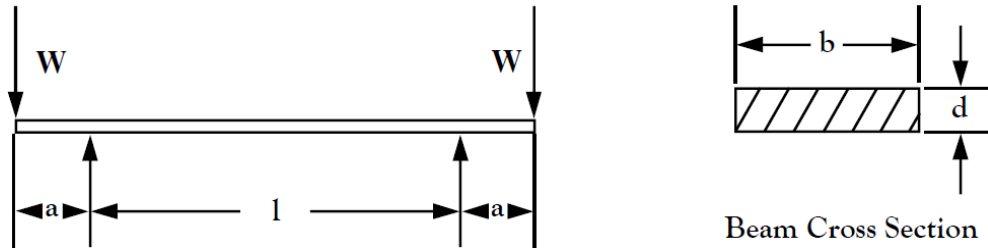
Test to be conducted

Carry out load /deflection tests, for beams of the same material as follows:

1. Test (ii) $l = 400\text{mm}$ $b = 12.7\text{mm}$ $d = 6.4\text{mm}$
 2. Test (iii) $l = 400\text{mm}$ $b = 25.4\text{mm}$ $d = 3.2\text{mm}$
 3. Test (iv) $l = 400\text{mm}$ $b = 25.4\text{mm}$ $d = 6.4\text{mm}$
 4. Test (i) $l = 800\text{mm}$ $b = 25.4\text{mm}$ $d = 6.4\text{mm}$
- ✓ For the effect of decreasing l compare tests (i) and (iv)
 - ✓ For the effect of increasing b compare tests (ii) and (iv)
 - ✓ For the effect of increasing d compare tests (iii) and (iv)

4.2.3. Experiment 3

To investigate, for a simply supported beam subjected to a uniform bending moment, the effect of variations in length over which the beam is supported.



Carry out load /deflection tests, for beams of the same material and cross-sectional

Dimensions as follows:

1. Test (i) $l = 700\text{mm}$ $a = 50\text{mm}$
2. Test (ii) $l = 600\text{mm}$ $a = 100\text{mm}$
3. Test (iii) $l = 500\text{mm}$ $a = 150\text{mm}$
4. Test (iv) $l = 400\text{mm}$ $a = 200\text{mm}$
5. In all tests $b = 25.4\text{mm}$ and $d = 6.4\text{mm}$

Specimen graphs for these experiments are included at the end of this manual. For more advanced students, surface strain measurements could be taken by attaching some form of strain gauge to the surface of the beam. These strain values could then be compared with those predicted by using the theory of bending.

4.3. Specimens I

4.3.1. Experiment 1

Simply Supported Beam (Central Point Load) Effect of

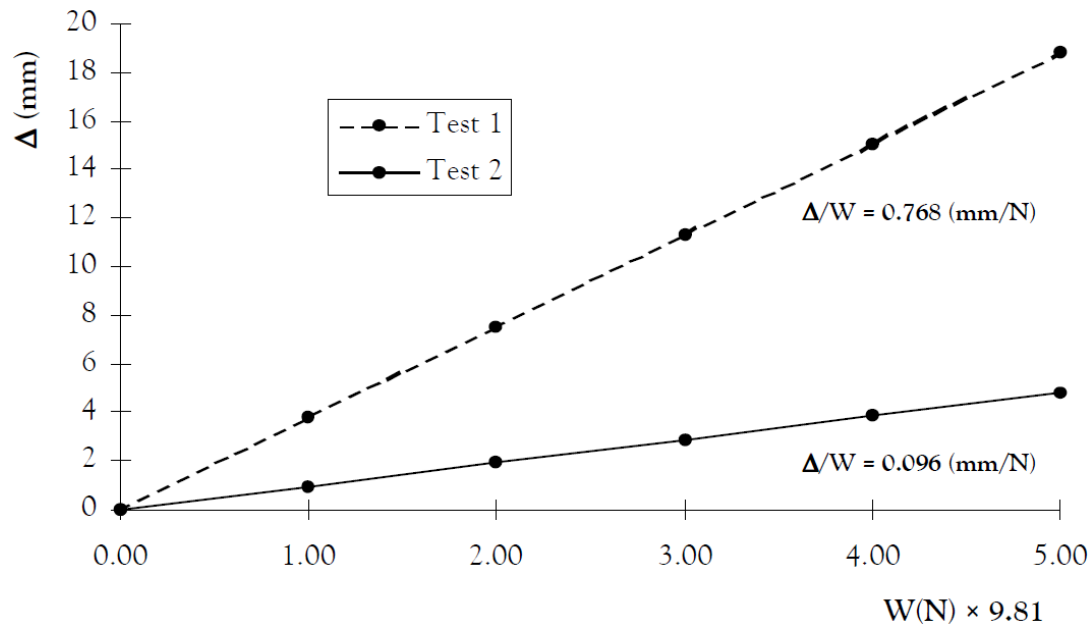


Figure 8: test results of sample 1 & 2

4.3.2. Experiment 1

Simply Supported Beam (Central Point Load) Effect of

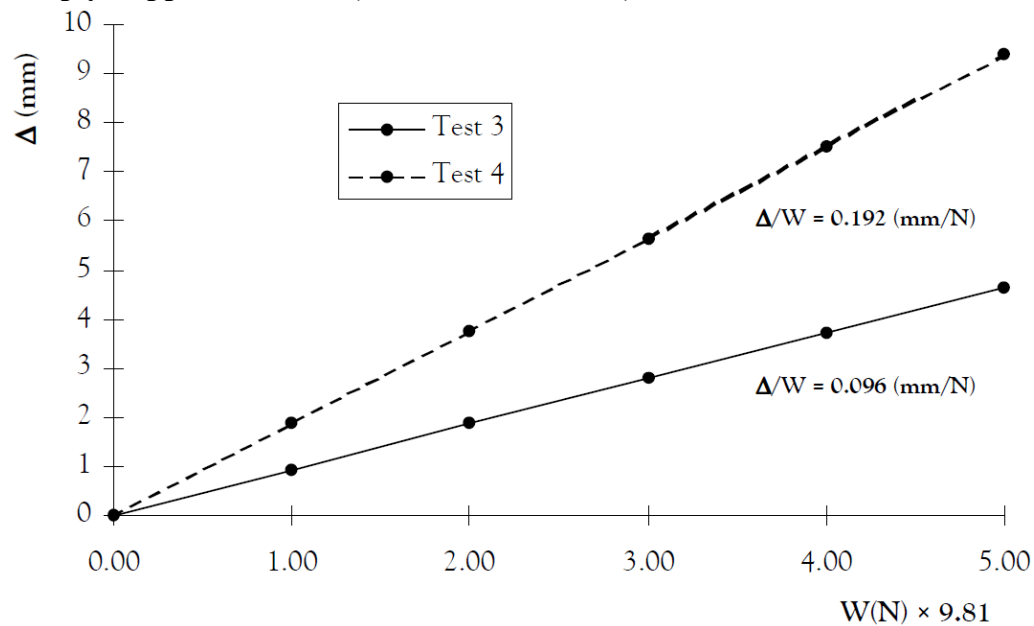


Figure 9: test results of sample 3 & 4

4.4. Specimen II

4.4.1. Experiment 1

Simply Supported Beam (Central Point Load) Effect of

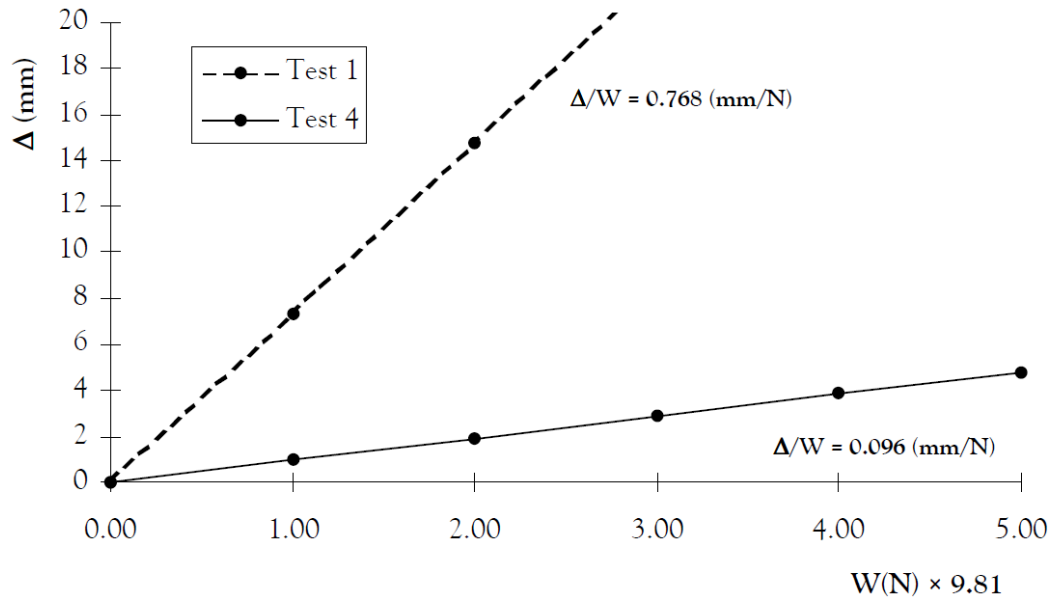


Figure 10: test results of sample 1 & 4

4.4.2. Experiment 2

Cantilever Beam (End Point Load) Effect of Decreasing I

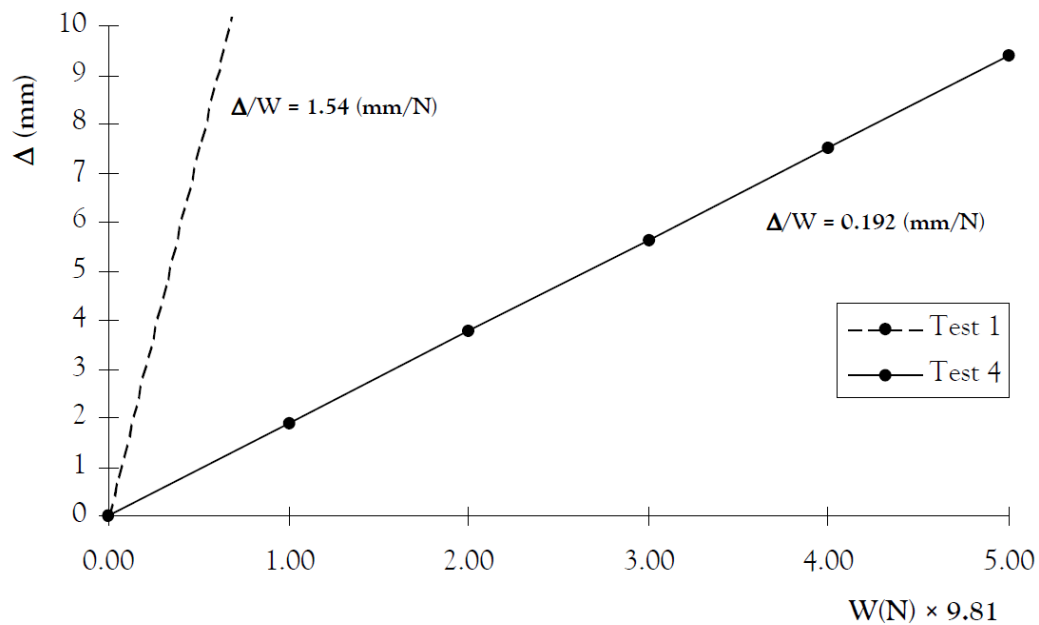


Figure 11: test results of sample 1 & 4

4.5. Specimen III

4.5.1. Experiment 2

Cantilever Beam (End Point Load) Effect of Increasing “b”

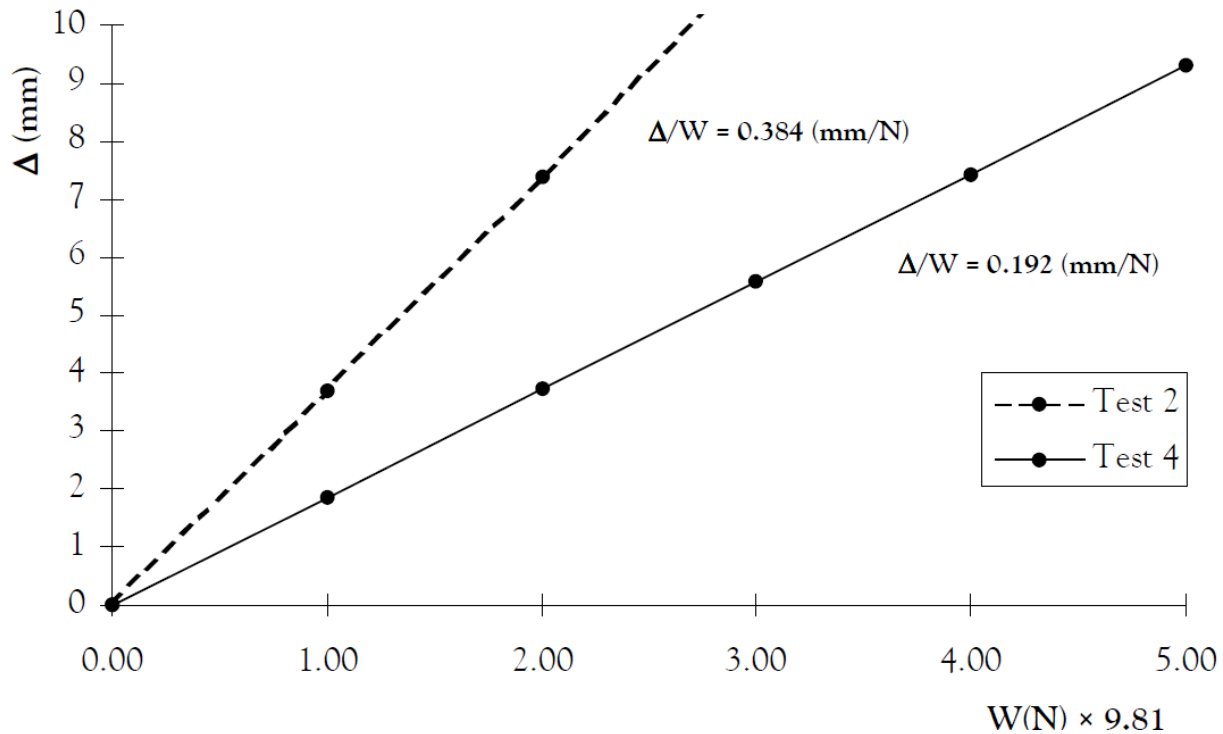


Figure 12: test results of sample 2 & 4

4.5.2. Experiment 2

Cantilever Beam (End Point Load) Effect of Increasing “d”

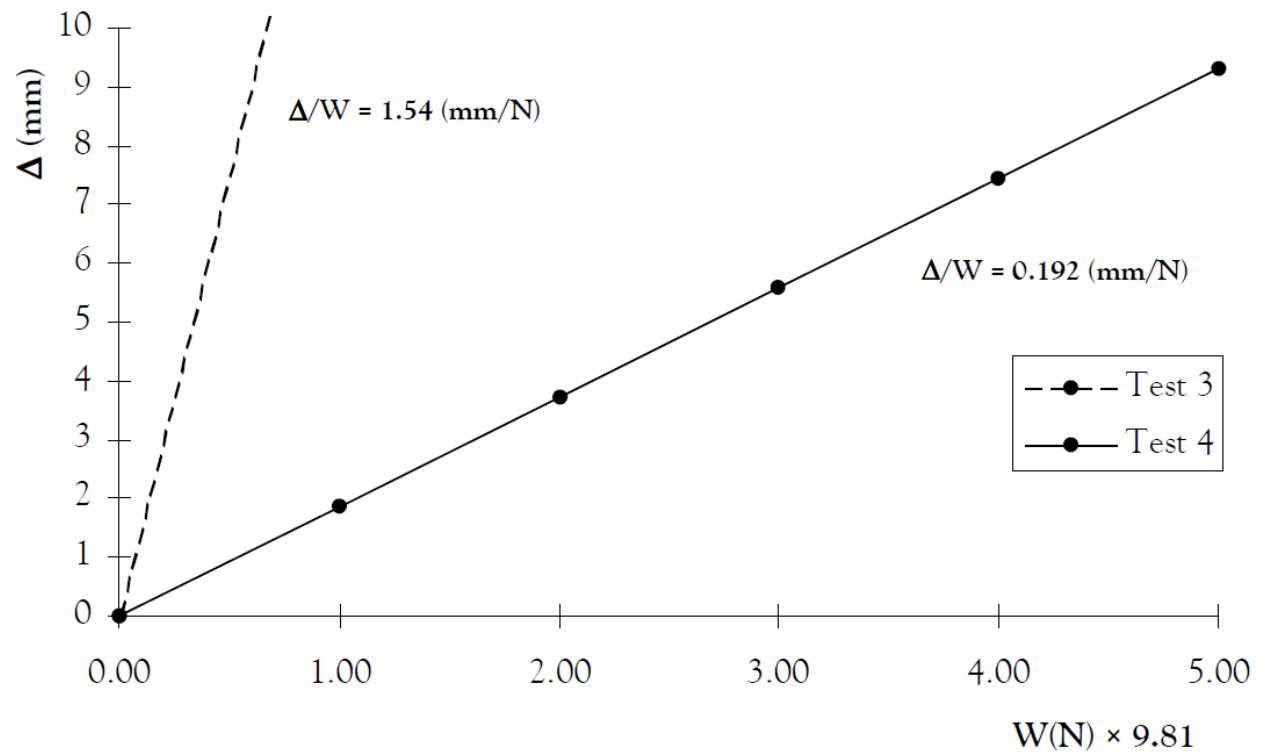


Figure 13: test results of sample 3 & 4

4.6. Specimen IV

4.6.1. Experiment 3

Simply Supported Beam (Uniform M)

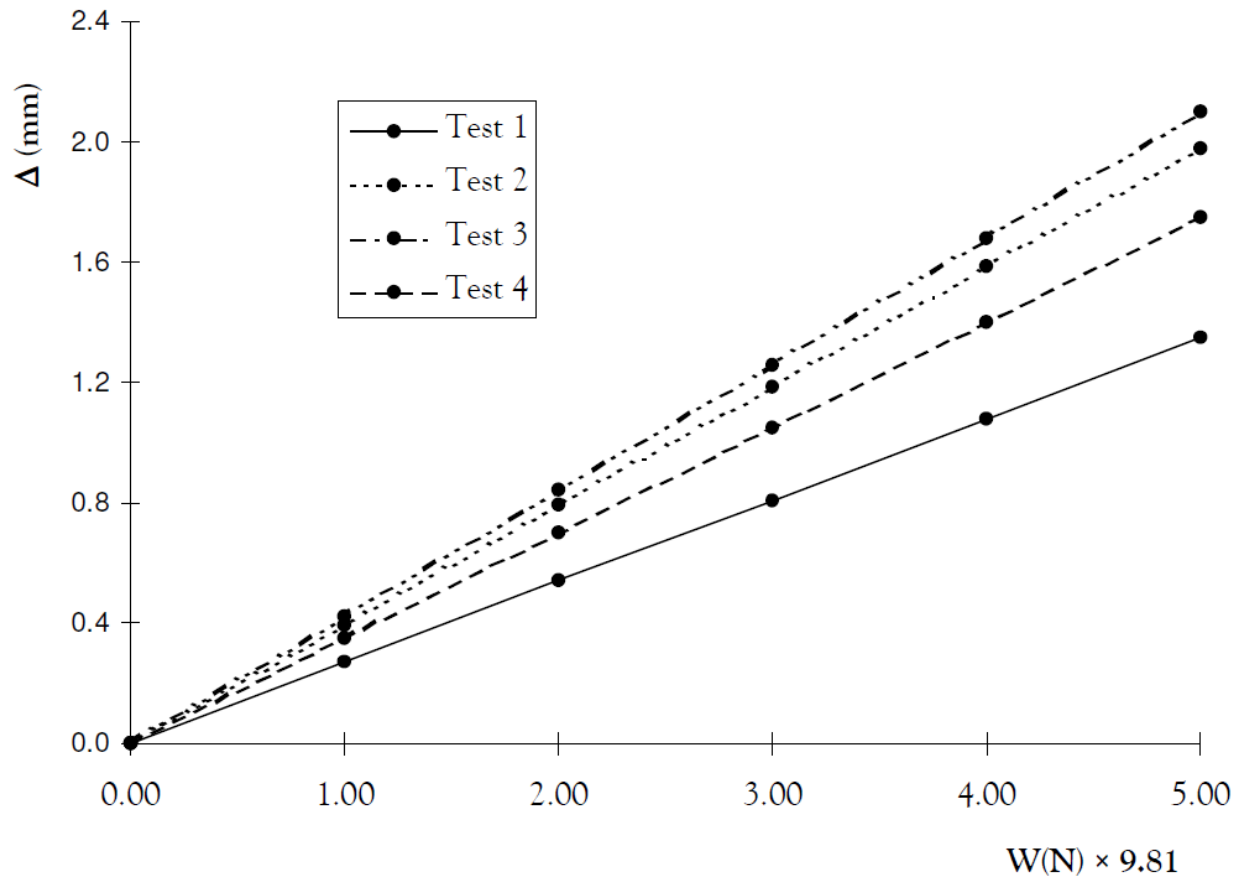


Figure 14: test results of sample 1, 2, 3 & 4

1. Test (i) $D/W = 0.0276$ (mm/N)
2. Test (ii) $D/W = 0.0405$ (mm/N)
3. Test (iii) $D/W = 0.0421$ (mm/N)
4. Test (iv) $D/W = 0.036$ (mm/N)

5. Torsion experiment

5.1. Introduction

This simple piece of apparatus has been designed for student laboratory exercises to investigate the elastic torsion characteristics of circular bars. Torque is applied to the specimen bars by means of a spring balance and torque arm. A dial gauge in contact with the torque arm enables the angle of twist to be measured. Matching bars in Steel, Brass, and Aluminum are supplied for direct comparison. This range of experiments includes:

5.2. Objective

1. To determine experimentally the relationship between the tensional moment and the angle of twist of a shaft.
2. To determine experimentally the relationship between the clamping length and the angle of twist.
3. To determine experimentally the value of the shear modulus of elasticity for steel, brass, and aluminum.

5.3. Description of equipment

Torsion of bars apparatus

Overall Dimension

Height: 0.52 m, Depth: 0.25m, Net weight: 9.5 Kg , Width: 0.50 m

5.4. Theory

1. The shaft is straight and of uniform circular cross-section throughout its length.
2. The applied torque is constant over its length and acts about the polar axis.
3. The induced stresses do not exceed the limit of proportionality.
4. Cross-sectional planes remain plane after strain.

$$\frac{T}{J} = \frac{\tau}{R} = \frac{G\theta}{L}$$

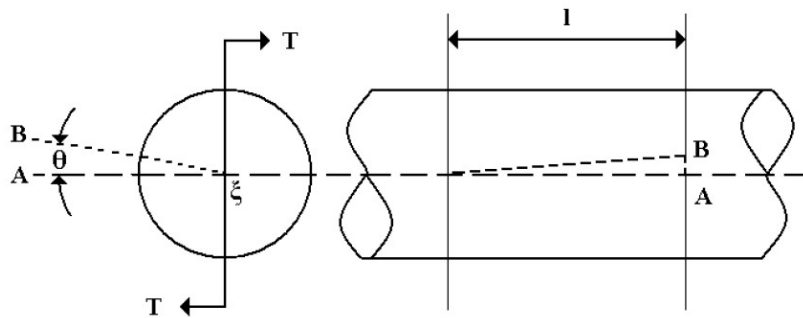
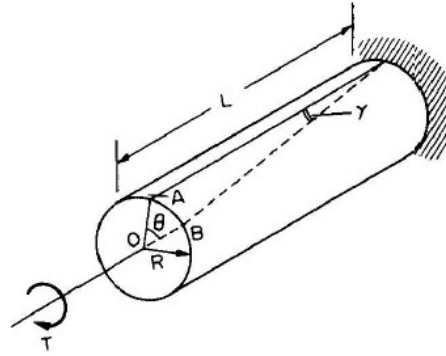


Figure 15: components of torsion

Let

θ = Angle of twist over a length l

T = the applied torsional moment

G = Shear modulus of elasticity

I = Polar second moment of area

It can be shown that
$$\frac{T}{I_p} = \frac{G\theta}{l}$$

$\therefore \theta = kt$ where $k = \frac{l}{GI_p}$ 1

$\theta = kl$ where $k = \frac{T}{GI_p}$ 2

$G = K \frac{T}{\theta}$ where $k = \frac{l}{I_p}$ 3

5.5. Operating the Equipment

1. Release clamp (1) and set the position of block (2) so that length l is the required specimen test length.
 2. Place specimen through chucks and clamp fixed chuck (4) with the selected groove set to the inner face of the chuck.
 3. Rotate hand wheel (7) to lower spring balance (8) to allow torque arm (6) to rest freely in the lower limit position.
 4. Clamp specimen in the free chuck (5) checking that the groove in the specimen is set to the inner face of the chuck.
 5. Raise the spring balance until contact is made with the torque arm. Check that the spring balance is reading zero, use nut (9) to adjust if necessary.
 6. Position dial indicator (10) to make contact with torque arm (6) at the position indicated by a groove.
 7. Set indicator to zero.
 8. Apply a small torque to the specimen by rotating the hand wheel (7) to raise the spring balance.
 9. Note the dial indicator reading.
 10. Increase the torque by equal increments and note the indicator reading for each applied torque.
 11. To vary the specimen test length l , lower the spring balance to the free, no-load position and release chuck (4) and clamp (1).
 12. Position block (2) with the inner face of the chuck (4) set to the desired groove in the specimen.
 13. Secure block (2) and clamp specimen in the chuck (4).
 14. Repeat the loading procedure.
- ❖ Where necessary, refer to the drawings in the Equipment Diagrams section.

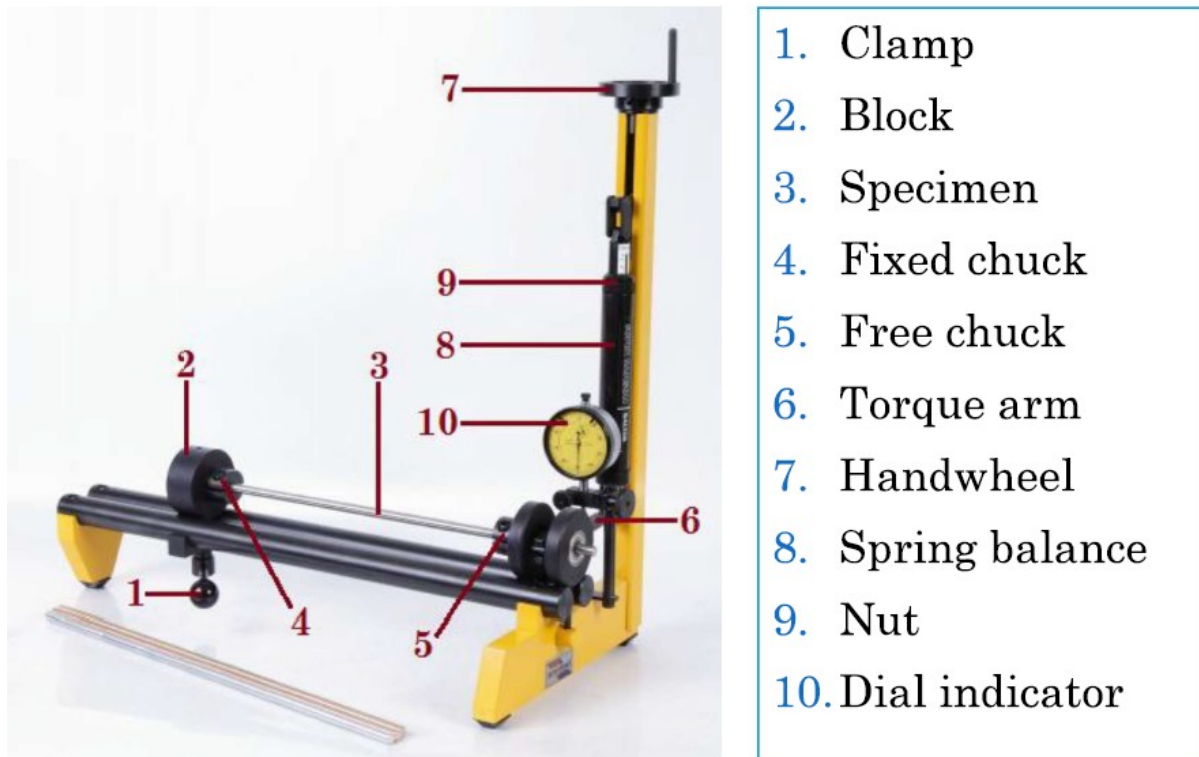


Figure 16: torsion of beam equipment

5.6. Equipment Specifications

Dimensions of Shafts

$d = 8\text{mm}$,

$l = 350\text{ mm (max)}$

Materials of Shafts

Steel, Brass, Aluminum

Environmental Conditions

This equipment has been designed for operation in the following environmental conditions. Operation outside of these conditions may result in reduced performance, damage to the equipment or hazard to the operator.

5.7. Laboratory Exercises

Index to Exercises

Exercise A - Torsional Moment v Angle of Twist

Exercise B - Clamping Length v Angle of Twist

Exercise C - Shear Modulus of Elasticity

5.7.1. Exercise A - Torsional Moment v Angle of Twist**Objective**

To investigate the relationship between torsional moment and angle of twist of a shaft.

Procedure

Place a specimen through the chucks and set the test length l at say 300mm. Clamp specimen (refer to procedure notes). Vary the torsional moment as indicated in the table and record the dial gauge Reading for each applied moment

The specimen is Steel.

Note: Torsional moment T (Nm) = Spring Balance (Kg) $\times$ 0.1g

Angle of twist (rad) = Dial Indicator (mm) $\times$ 0.02

Load (kg)	Dial Indicator(mm)	Dial Indicator(Nm)	Angle of Twist θ (Rad)
0.5			
1.0			
1.5			
2.0			
2.5			
3.0			
3.5			
4.0			

Plot the values on the graph.

$$\theta = T \cdot \frac{l}{G \cdot I_p}$$

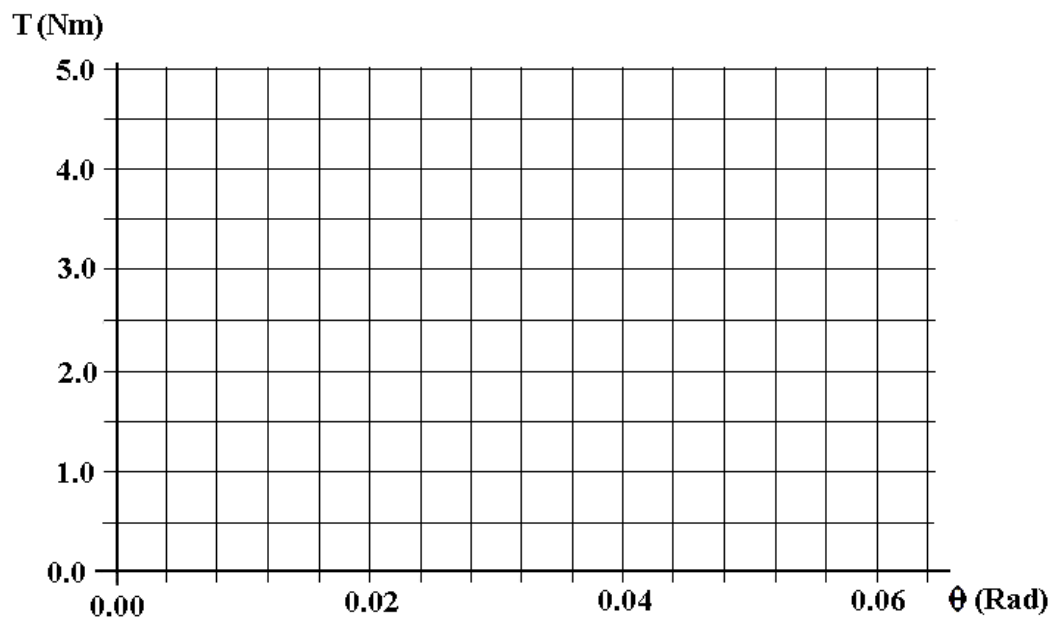


Figure 17: torsion Vs angle of twist graph

5.7.2. Exercise B - Clamping Length v Angle of Twist

Objective

To investigate the relationship between the clamping length and the angle of twist of a shaft.

Procedure

Place a specimen through the chucks and set the test length l at 350 mm. Clamp specimen (refer to procedure notes). Apply a torsional moment to the specimen (e.g: 3.0 kg on the spring balance) and record the dial indicator reading. Repeat with specimen clamped at varying lengths.

Specimen: Steel

Note: Torsional moment T (Nm) = Spring Balance (Kg) $\times$ 0.1g

Angle of twist (rad) = Dial Indicator (mm) $\times$ 0.02

Table 2: input and output results of length and angle of twist

Length l (mm)	Dial Indicator(mm)	Torque T (Nm)	Angle of Twist θ (Rad)
350			
300			
250			
200			
150			
350			

$$\theta = l \cdot \frac{T}{GJ_p}$$

Plot the values in the graph.

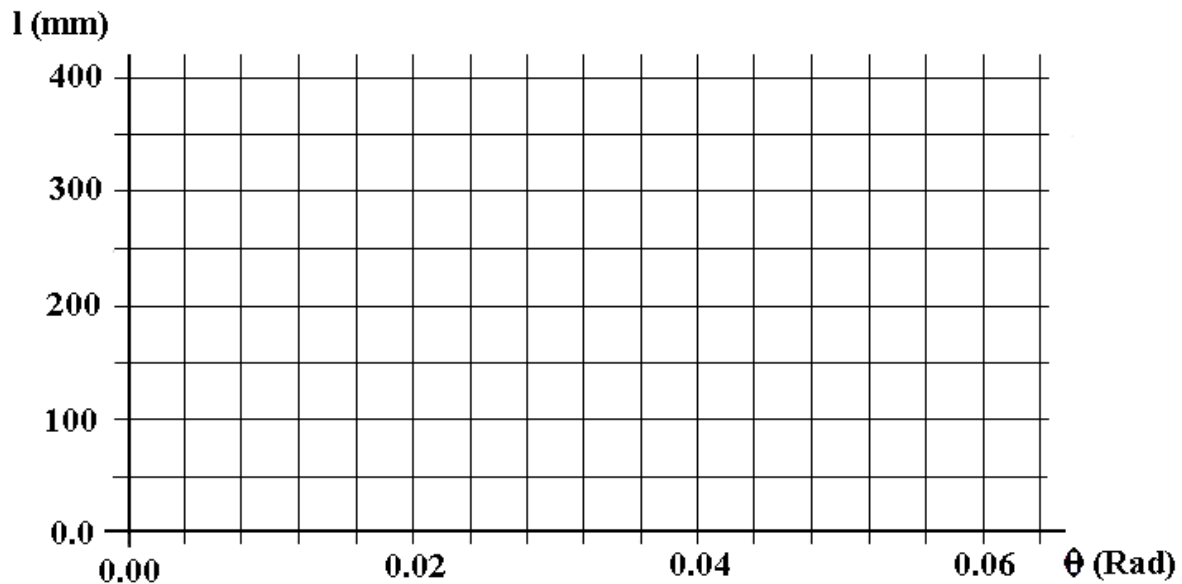


Figure 18: length Vs angle of twist graph

5.7.3. Exercise C - Shear Modulus of Elasticity

Objective

To determine the value of the shear modulus of elasticity for steel, brass, and aluminum.

Procedure

Place the steel test specimen through the chucks and set the test length l at 300 mm. Clamp specimen (refer to procedure notes). Vary the torsional moment as indicated in the table and record the dial indicator reading for each applied moment. Repeat with brass and aluminum specimens.

Table 3: : input and output results of material and angle of twist

Material	Spring Balance(kg)	Dial Indicator(mm)	Torque (Nm)	Angle of Twist (Rad)
Steel	1.0			
	2.0			
	3.0			
Brass	1.0			
	2.0			
	3.0			
Aluminum	1.0			
	2.0			
	3.0			

Note: Torsional moment T (Nm) = Spring Balance (Kg) $\times$ 0.1g

Angle of twist (rad) = Dial Indicator (mm) $\times$ 0.02

$$I_{\phi} = \frac{\pi d^4}{32}$$

$$d = 0.008 \text{ m}$$

$$l = 0.3 \text{ m}$$

1

Calculate the average values of $\frac{T}{\theta}$ for each specimen and use these values to

Determine the shear modulus of elasticity.

$$G = \frac{l}{I_{\phi}} \cdot \frac{T}{\theta}$$

2

Table 4: input and output results of material and shear modules

Material	$\frac{T}{\theta}$ (average)	$\frac{l}{I_{\epsilon}}$	G(N/m ²)
Steel			
Brass			
Aluminum			

5. Strut Buckling Unit

5.1 Introduction

Buckling is a phenomenon of structural instability that is generally suffered by slender structures under compression. A perturbing force acts over them and causes a failure on them before reaching the compressive strength limit. Buckling is evident because the piece is not straight anymore; it adopts a curved shape, known as the elastic curve. The appearance of bending in the system severely limits the compressive strength of a pillar or column, where a situation of elastic instability can be generated from a specific value of compression axial load, called critical buckling load. Deformation will increase till the failure of the structure. Bars or columns must be long enough, compared to their thickness (slenderness), to result in buckling. Besides, different factors, such as the materials used, the types of loads, ties or supports, are considered. The MFLT is a unit designed and manufactured by EDIBON to demonstrate the buckling of compressed structures, which allows the calculation of the critical buckling load in function of the geometry and/or supports.

5.2 Theory

When on a column or strut, whose length is several times larger than the characteristic measurement of the axial section, a compression force smaller than the compressive strength limit is applied; a side bending may appear, lead to the failure of the structure and make the element break by buckling due to its quick growth and the total loss of stability.

The first problems of stability in structures related to the buckling were solved by Euler. The problem posed by Euler started from the following prerequisites:

- Bar made of a perfectly elastic and homogeneous material, that is to say, which fulfills Hooke's Law.
- Completely straight bar.
- Centered load.
- Perfect connections, without friction.

Under these conditions, the vertical bar has a balanced position that will be studied.

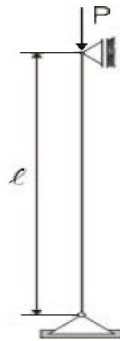


Figure 19: Balanced bar

Where:

P Load applied to the system

L Length

To determine the stability, the study starts by applying an infinitesimal horizontal force to the bar. As a consequence, the bar adopts another curved balance configuration, as the one indicated in figure 3.

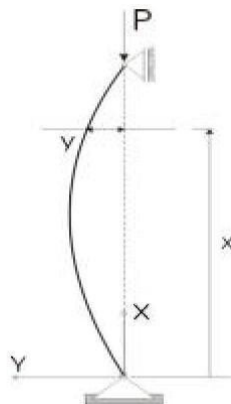


Figure 20: Bar with curved balance position

Where:

P	Load applied to the system
X	Length
Y	Side displacement

For a generic section located in an abscissa “x”, the bar has a “y” shift. If we state the balance between the external moment and the internal elastic moment, we will have:

$$P \cdot y = -E \cdot I \cdot y''$$

Equation 2: Equilibrium between the external moment and the internal elastic moment

Where:

P	Load applied to the system
y	Side displacement
E	Young's modulus
I	Area moment of inertia

Expand the second order differential equation to obtain the homogeneous

Equation:

$$P \cdot y + E \cdot I \cdot y'' = 0$$

Equation 3: Homogeneous second order differential equation

Grouping terms to facilitate the solution of the differential equation and naming, $\alpha^2 = \frac{P}{E \cdot I}$, we obtain: $y'' + \alpha^2 \cdot y = 0$

Equation 4: Simplified homogeneous second order differential equation

To solve this equation a general solution must be set out:

$$y = C_1 \cdot \sin(\alpha \cdot X) + C_2 \cdot \cos(\alpha \cdot X)$$

Equation 5: General solution for the ordinary differential equation (ODE)

To obtain a particular solution, the value of the constants C_1 and C_2 must be obtained by applying the boundary conditions:

- $X=0; Y=0$:

$$0 = C_1 \cdot \sin(\alpha \cdot 0) + C_2 \cdot \cos(\alpha \cdot 0)$$

$$0 = C_2$$

- $X=L; Y=0$:

Knowing that $C_2=0$ and replacing in the previous equation:

$$0 = C_1 \cdot \sin(\alpha \cdot L)$$

Two situations can take place to fulfill the nullity of the previous equation:

- 1) $C_1=0$:

In this case, the elasticity equation would be null and would be representing the initial vertical configuration of the bar. Logically, this case does not interest us, since we are looking for other balance configurations.

- 2) $\sin(\alpha \cdot L)=0$:

$$\alpha \cdot L = n \cdot \pi$$

Solving for α , squaring and equaling with the expression used in equation 4,

we obtain:

$$\frac{n^2 \cdot \pi^2}{L^2} = \frac{P}{E \cdot I}$$

Solving for the load (P) we obtain this equation:

$$P = \frac{E \cdot I \cdot n^2 \cdot \pi^2}{L^2}$$

Equation 6: Expression to calculate the load applied

For values of the load P that verify the previous equation, different elasticity values are obtained which correspond to balance configurations of the bar. The smaller load of all generated by this situation corresponds to $n = 1$. Such load is the “critical load”.

$$P_{crit} = \frac{E \cdot I \cdot \pi^2}{L^2}$$

Equation 7: Expression to calculate the critical load of a bar

P_{crit}	Critical load supported by the system	N
E	Young's modulus	N/m ²
I	Area moment of inertia	M^4
L	Length	m

Where:

Due to the geometry of the bars under study, the area moment of inertia corresponds to a rectangular flat figure, thus:

$$I = \frac{1}{12} \cdot b \cdot h^3$$

Equation 8: Area moment of inertia for rectangular profile structures

Where:

I	Area moment of inertia	M^4
b	Width of the specimen	m
h	Thickness of the specimen	m

Equation 7 is only valid when deformations are small. Thus, we may state that the solution found for the critical load is the limit of the loads P when the curved balance configuration is as close to the vertical as desired. Finally, we can analyze the meaning of the critical loads corresponding to $n = 2, 3, 4$, etc. If we consider, for example, the case of $n = 2$, we can see that $P = 2 \cdot P_{crit}$ and the undetermined sinusoidal elasticity coefficient is formed by a double half-wave. This load has only theoretical interest. It corresponds to the critical load in case the bar was fixed in the center of its span by means of movable support.

Support Types

We have studied in the previous section the buckling of a bar hinged at both ends under certain hypotheses. It is also possible to carry out a similar study for other connection conditions, making it possible to establish the corresponding critical load for each case. Next, we are going to indicate the most common support cases.

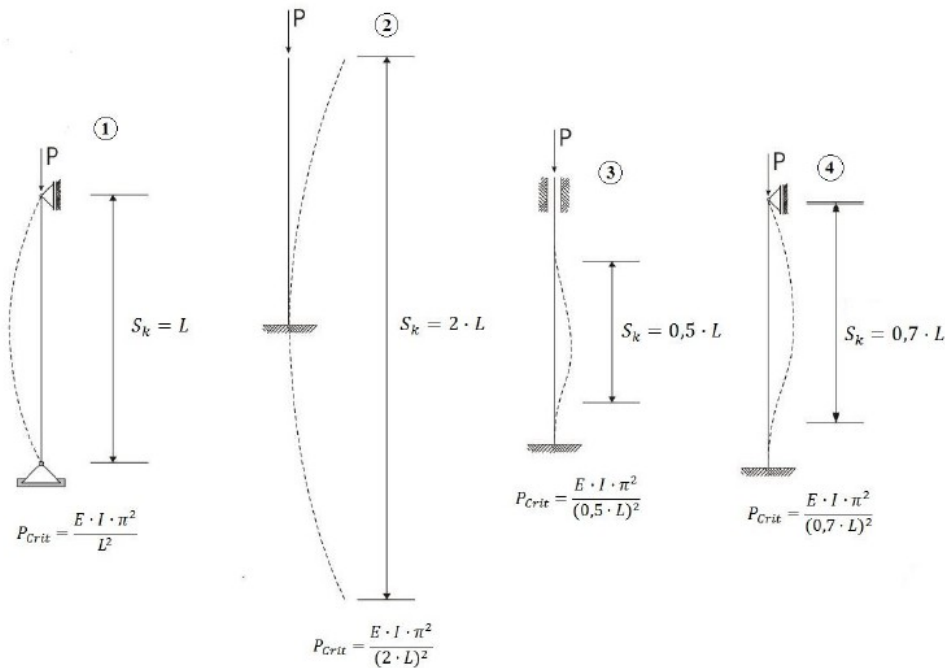


Figure 21: Other forms of support

Where:

Figure 1: Configuration of a bar hinged at both ends.

Figure 2: Configuration of a bar free at one end and fixed at the other.

Figure 3: Configuration of a bar fixed at both ends.

Figure 4: Configuration of a bar hinged at one end and fixed at the other.

Observing the figure 4, we can check that the expressions that correspond to the critical loads for the different types of support are very similar to those of the bar hinged at both ends, varying only in one constant. From a practical point of view, it is convenient to be able to cover any support case with only one expression for the critical load. This is achieved by transforming the piece into a fictitious bar hinged at both ends with an ideal span that depends on the real span and on the real connection conditions. This fictitious span is called “buckling span” or “buckling length” and is represented by S_K . Therefore, a generic expression can be defined to calculate the critical load regardless of the supports of the bar under study.

$$P_{crit} = \pi^2 \cdot \frac{E \cdot I}{S_K^2}$$

Equation 9: Generic expression to calculate the critical load

Where:

P_{crit}	Critical load supported by the system	N
E	Young's modulus	N/m ²
I	Area moment of inertia	M^4
SK	Buckling length	m

Taking into account the previous development, the values of the buckling length in the function of the different supports can be obtained, thus:

Buckling length of bar hinged at both ends	SK = L
Buckling length of bar free at one end and fixed at the other	SK = 2 · L
Buckling length of bar fixed at both ends	SK = 0,5 · L
Buckling length of bar hinged at one end and fixed at the other	SK = 0,7 · L

There are some tables where the corresponding buckling length can be determined for other structural elements, such as the columns of a portico or bars with a variable section.

5.3. Description of the unit

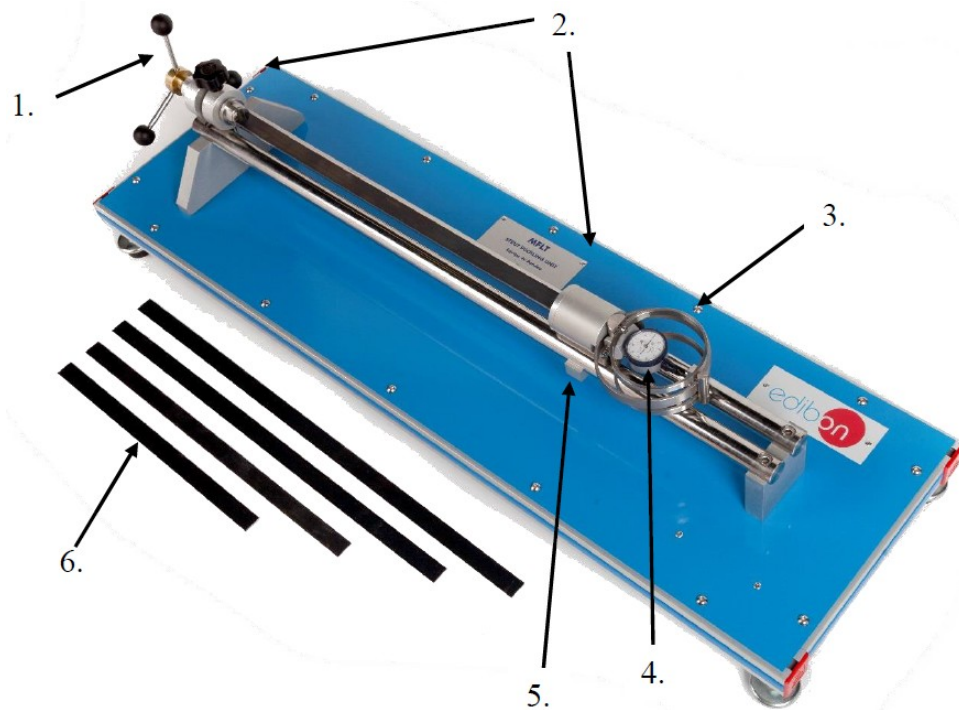


Figure 22: Strut buckling unit

1. TA-1: fine pitch fixing nut for the system compression.
2. SO-1: bars support to carry out the experiments. These elements enable the study of a fixed or hinged system.
3. MU-1: metal elastic ring. It allows for obtaining the load applied to the system.
4. RC-1: dial gauge. It measures the deformation of the elastic ring which, together with the spring constant (refer to the specifications section), allows for calculating the load applied to the system.
5. MO-1: clamp. Movable element to fix the distance at which the test is done.
6. SB-1: the set of bars. Group of specimens of the different sizes used in the experiments.

5.3.1. Description of the process

The strut buckling unit, made to demonstrate the buckling in structures, is a bench-top unit whose frame is made of anodized aluminum profiles. The unit consists of two guide bars along which the fastening clamps of the test bars move. The test bar under study must be located on the supports (SO-1), fixing the distance with the clamp (MO-1), to carry out the buckling tests. In order to carry out the practical exercises correctly, it is very important to apply the force with only one axial component. For that purpose, the MFLT unit is provided with a fine pitch fixing nut (TA-1) to apply the load to the system. The MFLT unit has a system to measure the compressive force exerted on the structure composed by an elastic ring (MU-1), which is deformed when a load is applied to the system, and a dial gauge (RC-1), which indicates that deformation. The compressive force will be obtained by Hooke's law, where the deformation is the value registered by RC-1 and the spring constant is defined in the specifications section.

A wide range of test bars is supplied together with the MFLT unit. They have different lengths but the same profile. Different end conditions can be applied to them by using the clamps correctly.

5.3.2 Practical possibilities

1. Determination of the spring constant to calculate the stress applied to the system.
2. Determination of the critical buckling load of a rectangular horizontal bar fixed at both ends.
3. Determination of the critical buckling load of a rectangular horizontal bar fixed at one end and hinged at the other end.
4. Determination of the critical buckling load of a rectangular horizontal bar hinged at both ends

Other practical possibilities:

1. Calibration of the metal elastic ring using a compression dynamometer.

5.3.3 Specifications

The strut buckling unit is mounted on a frame made of anodized aluminum profiles and a steel plate painted in epoxy.

The strut buckling unit is basically composed of:

- Two guides made of stainless steel to stroke along with the movable clamp (MO-1). They allow the testing of bars up to 650 mm long.

- Clamp (MO-1) consisting of the system to measure the load applied to the unit (dial gauge (RC-1) and elastic metal ring (MU-1)) and a lower handle to fix the distance required for the study.
- Dial indicator (RC-1) with a measuring range of 0 – 5 mm and a precision of 0.01 mm. It is used to determine the compression to which the test bar is subjected as a load is applied.
- Metal ring (MU-1) made of steel (with a certain degree of elasticity). It is used to obtain the load of the system by applying Hooke's law together with the measurement obtained with RC-1.

$$F = K \cdot X$$

Equation 1: Hooke's law formula

Where:

F	Force	Kg
K	Spring constant	Kg/mm
X	Displacement	mm

The generic value of the spring constant is 202 Kg/mm, but it can vary depending on the unit. For more accurate results, we recommend using the value obtained in the practical exercise 1 as the real value of that constant

- Nine test bars made of tempered steel. They are of different lengths and measure 19mm width per 1.5mm thick.

5.3.4. Dimensions and weight

- Dimensions: 100 x 30 x 22 cm approx.
- Volume: 0.066 m³.
- Weight: 10 kg.

5.3.5. Required services

- The strut buckling unit requires a leveled workbench to perform the test in optimal conditions.

5.3.6. Recommended accessories (not included in the unit)

- To calibrate the elastic metal ring with a compression dynamometer (practical exercise NOT included in this manual), the user must buy one of the compression dynamometers found in the market to carry out the practical exercise reliably.
- Set of weights “C type” manufactured by Edibon to calibrate the elastic ring (refer to the practical exercise 1 in the laboratory practical exercises section).

5.4. Operations

5.4.1 Mounting the clamp to calibrate the elastic ring

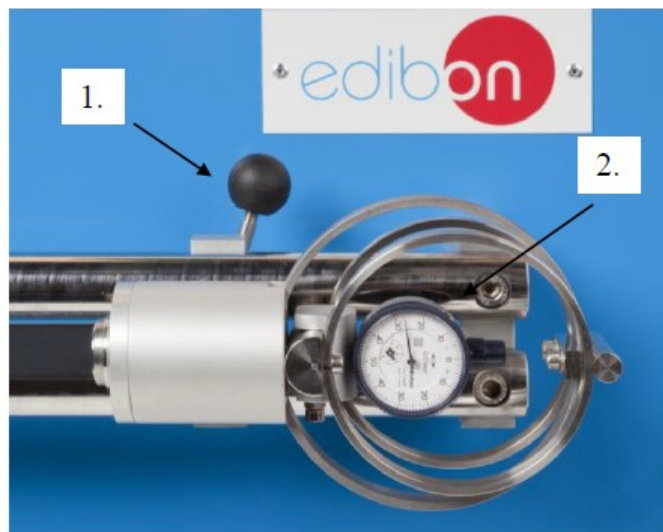


Figure 23: System to calculate the load applied

The elastic ring (MU-1) must be disassembled from the unit to be calibrated. To that end, the clamp (MO-1) has a handle (1) (at the bottom) to fix its position. The clamp is released by turning the handle anticlockwise.

When the clamp is disassembled, place it in a vertical position to put the weight on it. The dial gauge must be calibrated before applying any load, to that end, the dial gauge (RC-1) includes a movable crown (2) to displace the scale. To calibrate the dial gauge (RC-1) turn the crown till the zero in the scale coincides with the needle of the dial gauge.

After completing the previous steps, the elastic ring (MU-1) can be calibrated. To that end, different weights are gradually located and the deformation generated is recorded for the later calculation of the constant

5.4.2 Positioning the ends to perform the tests under different support conditions

To carry out the tests under fixed, hinged or combined conditions the MFLT unit has two supports (SO-1) to modify those conditions. Both supports are equal and have a groove where the bar that simulates the fixed-end condition is introduced. The pieces are beveled to simulate the support.

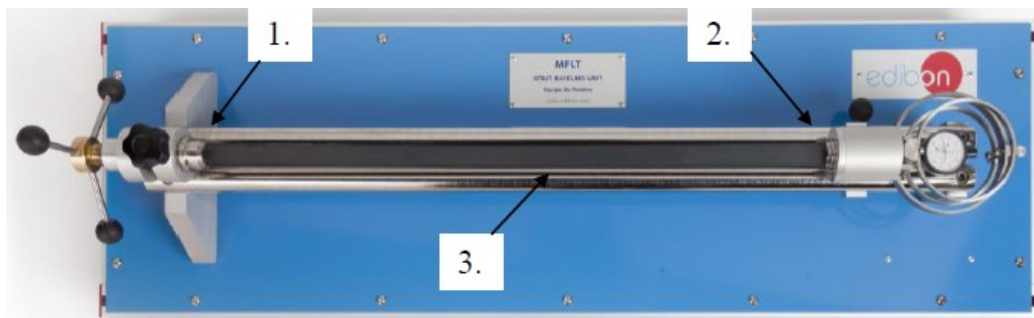


Figure 24L MFLT unit

The position of the supports must be adjusted to work under different conditions. The support (1) is removable and has two tracks that attach the piece to the unit and a screw with the knob, which works as a guide, to prevent the piece from rotating during the test.

When removing the support (1), it is advisable to select the configuration that makes the tested bar bend laterally, thus preventing the bar from resting on the stainless steel guides (3) when bending. Regarding the support (2), it is not a removable piece but it can rotate, enabling the user to select the desired configuration

5.4.3 Adjusting the distance of the clamp

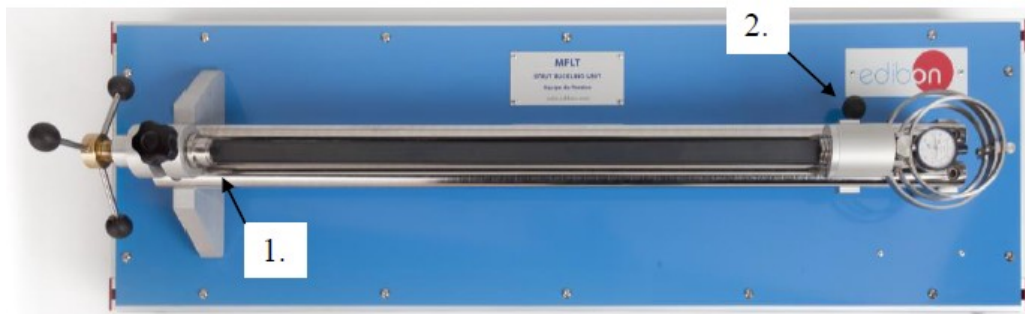


Figure 25: MFLT unit

To adjust the distance of the clamp it is necessary to use the bar under study. To that end, the bar is located on the support (1) with the desired conditions and, being the clamp released (2), it is displaced to the appropriate point. When it is located at a suitable distance, block the clamp again with the handle (2).

It is recommended to make sure that the bar is secured and the clamp does not move to prevent possible accidents or the obtaining of distorted data.

5.4.4 Load application and test completion

The MFLT unit includes a fine pitch fixing nut (TA-1) to apply load to the system. To that end, the bar understudy must be located in the unit and the clamp (MO-1) is attached to that unit.

Having the unit ready, gradually rotate the fixing nut (TA-1) clockwise and record the values of the deformations registered by the dial gauge (RC-1). The test will have finished when the deformation registered by the dial gauge (RC-1) remains constant although the force is still being applied with the fixing nut (TA-1).

It is important to apply force in a gradual and slow manner. This way, the deformation generated will be registered more efficiently. Besides, it allows the user to observe the point when the critical load is reached.

Warning: do not exceed the critical load since plastic deformations can be generated in the test bars.

5.5. Main instructions, warnings, and precautions

Before starting the practical exercises with the MFLT unit, check that the following conditions are fulfilled:

- Verify that the work site where the MFLT is located is clean, dry and leveled.
- The use of safety goggles is advisable.
- The fixing nut must be lubricated, just pour a few drops of oil on it and screw it so the oil covers the entire nut.
- Do not apply load higher than the critical buckling load during the tests to avoid plastic deformation of the bars.
- To obtain correct experimental data the bar must be completely straight. To that end, straighten up the test bars and, if they are excessively curved, replace them by a new set.
- Due to the testing requirements, the unit includes some pieces where corrosion could appear. To prevent corrosion or reduce it, the pieces (elastic ring and test bars) should be always covered by an oil patina.

5.5.1. Laboratory practical exercises

It is advisable to read this manual carefully before carrying out the practical exercises. The procedure that you may follow to carry out the practical exercises with this unit is described below.

5.5.2 Practical exercise 1:

Determination of the spring constant to calculate the stress applied to the system

Objective

The objective of this practical exercise is to determine the constant of the elastic ring that will be used later on to calculate the force applied to the system.

Required elements

- The clamp of the strut buckling unit.
- Set C of weights (not included).

Experimental procedure

1. Refer to the operation section “Mounting the clamp to calibrate the elastic ring”.
2. Calibrate the dial gauge to zero to start the test.
3. Add weights to the clamp gradually and record load and deformation in the table.

4. Represent force versus deformation and obtain the value of the slope that will correspond to the spring constant value.

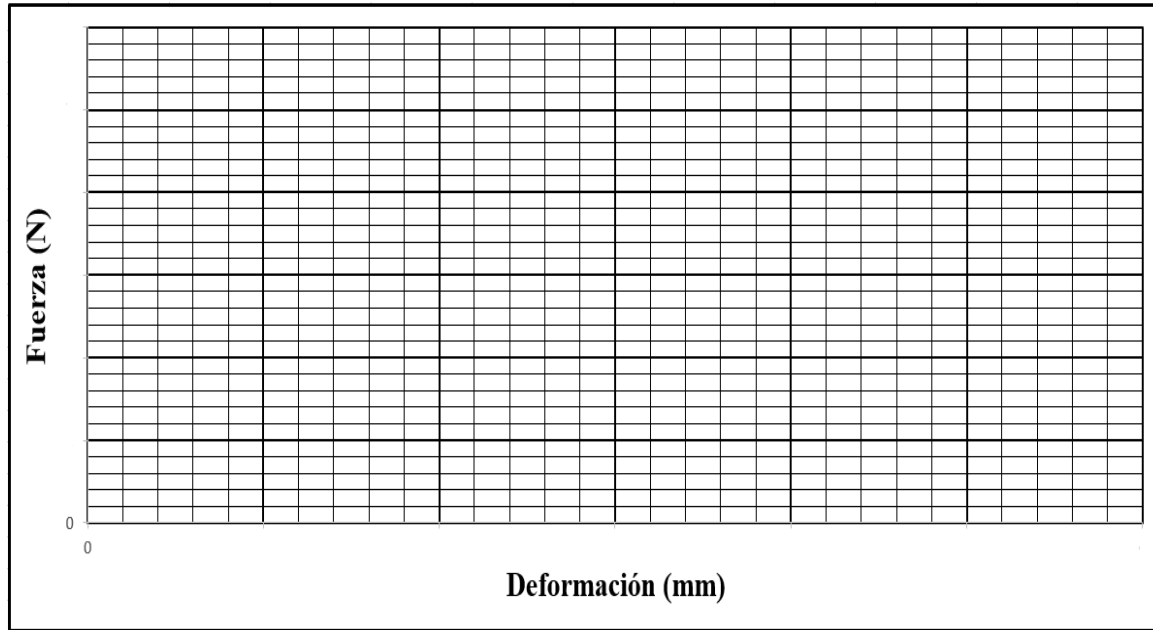
Worksheet

Student: _____ Year: _____

Exercise: _____

No	Force (N)	Deformation (mm)

Hooke's law: $F = K.X$



5.5.3 Practical exercise 2:

Determination of the critical buckling load of a rectangular horizontal bar fixed at both ends

Objective

The objective of this practical exercise is to calculate the critical buckling load in a bar fixed at both ends.

Required elements

- strut buckling unit.
- Set of bars supplied with the unit.

Experimental procedure

1. Place the ends as explained in the operation section “Positioning the ends to perform the tests under different support conditions”.
2. Adjust the distance of the clamp according to the operation section “Adjusting the distance of the clamp”.
3. Set the dial gauge to zero before performing the test.
4. Apply load gradually and record both the deformation and the applied load values (refer to the operation section “Load application and test completion”).
5. Repeat from step 2 on as many times as required to perform the rest of the tests with the bars of different lengths.

Worksheet

Student: _____ Year: _____

Exercise: _____

Length of the bar: _____ mm

Deformation (mm)											
Load (N)											

5.5.4 Practical exercise 3:

Determination of the critical buckling load of a rectangular horizontal bar fixed at one end and hinged at the other end

Objective

The objective of this practical exercise is to calculate the critical buckling load in a bar fixed at one end and hinged at the other end.

Required elements

- strut buckling unit.
- Set of bars supplied with the unit.

Experimental procedure

1. Place the ends as explained in the operation section “Positioning the ends to perform the tests under different support conditions”.
2. Adjust the distance of the clamp according to the operation section “Adjusting the distance of the clamp”.
3. Set the dial gauge to zero before performing the test.
4. Apply load gradually and record both the deformation and the applied load values (refer to the operation section “Load application and test completion”).
5. Repeat from step 2 on as many times as required to perform the rest of the tests with the bars of different lengths.

Worksheet

Student: _____ Year: _____

Exercise: _____

Length of the bar: _____ mm

Deformation (mm)											
Load (N)											

5.5.5 Practical exercise 4:

Determination of the critical buckling load of a rectangular horizontal bar hinged at both ends

Objective

The objective of this practical exercise is to calculate the critical buckling load in a bar hinged at both ends.

Required elements

- Strut buckling unit.
- Set of bars supplied with the unit.

Experimental procedure

1. Place the ends as explained in the operation section “Positioning the ends to perform the tests under different support conditions”.
2. Adjust the distance of the clamp according to the operation section “Adjusting the distance of the clamp”.
3. Set the dial gauge to zero before performing the test.
4. Apply load gradually and record both the deformation and the applied load values (refer to the operation section “Load application and test completion”).
5. Repeat from step 2 on as many times as required to perform the rest of the tests with the bars of different lengths.

Worksheet

Student: _____ Year: _____

Exercise: _____

Length of the bar: _____ mm

Deformation(mm)											
Load (N)											

6. Appendix:

Worksheet 1: shear and bending unit experiment data format

Inputs and results

No.	App Point	Load applied (W)	W1 (3) grams	W2 (4) grams

Worksheet 2:

Exercise A - Torsional Moment v Angle of Twist

Load (kg)	Dial Indicator(mm)	Dial Indicator(Nm)	Angle of Twist θ (Rad)
0.5			
1.0			
1.5			
2.0			
2.5			
3.0			
3.5			
4.0			

Exercise B - Clamping Length v Angle of Twist

Length (l)(mm)	Dial Indicator(mm)	Torque T (Nm)	Angle of Twist θ (Rad)
350			
300			
250			
200			
150			
350			

Exercise C - Shear Modulus of Elasticity

Material	Spring Balance(kg)	Dial Indicator(mm)	Torque (Nm)	Angle of Twist (Rad)
Steel	1.0			
	2.0			
	3.0			
Brass	1.0			
	2.0			
	3.0			
Aluminum	1.0			
	2.0			
	3.0			

: buckling experiment result

Worksheet 3

Buckling exercise 1:

No	Force (N)	Deformation (mm)

Buckling exercise 2:

Deformation (mm)											
Load (N)											

Buckling exercise 3:

Deformation (mm)											
Load (N)											

Buckling exercise 4:

Deformation(mm)											
Load (N)											

7. Reference

1. Edibon Engineering and technical teaching equipment manufacturing
2. Armfield engineering teaching and research Equipment
3. Strength of material books
4. Deferent lab Manual preparation website, MOS, MT lab manual preparation